

# Design and Development of Posture Detection System Using Load Sensor

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**Abstract** - A state of health, physically, mentally, spiritually, and socially, which is essential. Needed by the community in life to live productively socially and economically. One that affects health is sitting posture. Most of the time, humans spend their time sitting. The wrong sitting posture will significantly affect a person's health. Detecting poor posture can help prevent profound damage to human health. This thesis builds a sitting posture detection system using a force sensor (load sensor), which is useful for providing information on the user's pressure when sitting. This tool will get pressure from the hip seat. It can also provide ergonomic value to products such as Posture Cloud Medi Fit Stores. The results displayed by this tool are in the form of sit mapping. Users can draw conclusions about the correct way to sit and health diagnoses by knowing the pressure when sitting.

**Keywords:** Load Sensor, Arduino, Sit Mapping, Sitting Posture

## I. INTRODUCTION

According to YC Li [1], a person with poor sitting posture for an extended period will cause several problems, such as poor blood circulation. For paralyzed patients, poor sitting posture will cause problems with bedsores (Decubitus). According to Sun Bei [2], sitting for long periods with poor posture can harm physical and mental health. Poor sitting posture is a significant factor in myopia and cervical and lumbar vertebrae pain.

Detecting poor posture can help prevent profound damage to human health. In addition, recent studies have shown that jointly analyzing the relationship between sitting habits and learning efficiency is very important. For example, a stable sitting posture always represents a sculpture focusing on concentration [2].

This thesis builds a sitting posture detection system using a force sensor (load sensor) which helps provide information on the user's pressure when sitting. This tool will get pressure from the hip seat. It can also provide ergonomic value to products such as Medi Fit Store's Posture Cloud.

The results displayed by this tool are in the form of sit mapping so that users know how they feel when they sit down. Users can draw conclusions about the correct way to sit and health diagnoses by knowing the pressure when sitting.

## II. MATERIAL AND METHODS

Understanding the posture and movement of the human body has been a concern for a long time in the health field. This effort can articulate the patterns and parameters of all human activities, such as regular healthy sitting and deviations from a standard sitting characteristic of particular forms of clinical movement disorders, from neurological conditions to musculoskeletal conditions [3].

Nowadays people are used to receiving medical treatment in hospitals. The patient's family always chooses home care when the patient requires long-term care. Nevertheless, with limited human resources, nurses and doctors find it challenging to offer home care. Telehealth care that obtains patient information from devices and uses telecommunications networks will be a new healthcare method [1].

### A. Load Sensor/Force Sensor

A load sensor or load cell consists of a strain gauge or more, which is attached to a metal rod in the form of a ring. A load cell can adjust the number of strain gauges as needed. The strain gauge in the load cell is made of a foil grid.

The foil grid is a long thin wire arranged in a zigzag manner. The force sensor works as a force-sensing resistor in an electrical circuit. When the Force Sensor is not applied, the resistance is very high. When a force is applied to the sensor, this resistance decreases. Resistance can be measured with a multimeter, then apply a force to the sensing area [4].

### B. The Microcontroller

The microcontroller used in this tool is Arduino Nano. Arduino Nano is a small, easy-to-use board based on the ATmega328P architecture with an ATmega328 microcontroller operating at 5V. It uses 32 KB of flash memory, of which the bootloader uses 2 KB. Arduino Nano processor's clock speed is 16 MHz. Containing 16 analogs IN pins and 22 I/O pins, the PCB size is 18x45 mm and weighs 7 g [4].

### C. Bluetooth

The Bluetooth module used is Serial Bluetooth. Serial Bluetooth is a wireless communication module via Bluetooth which operates at a frequency of 2.4GHz with a choice of two connectivity modes. Mode 1 acts as an enslaved person or data receiver only, and mode 2 acts as a master or can act as a transceiver. The application of this component is very suitable for electronics projects with wireless or wireless communication.

The applications in question include control system applications, monitoring, or a combination. The interfaces used to access this module are serial TXD, RXD, VCC, and GND. Furthermore, there is an LED (built-in) indicator of Bluetooth connection to other devices, such as fellow modules, android smartphones, and so on.



Fig.1. Flow Diagram Research

#### E. Scales

The weighing module used is HX711, which has the working principle of converting the measured resistance changes into voltage quantities through an existing circuit. The module communicates with the microcontroller via TTL232.

#### F. Project Circuit Board (PCB)

PCB is a board used to connect electronic components with a layer of conductor paths. PCBs are divided into three types: PCBs that have one layer of copper (Single Sided), those that are layered with two layers of copper (double-sided), and PCBs that have several layers of copper or often called Multilayer PCBs.

#### G. System Design

This tool provides information on the user's sitting pressure on the hip seat. For how to use it, namely by occupying the device so that the sensor will provide a value to the server via Bluetooth, the diagnostic results will be obtained after the output value of the sensor is processed using a supporting application (MATLAB). The flow diagram(flowchart)of the research conducted is shown in Fig. 1.

#### F. PCB design

Here Ares Proteus showed in Fig. 2.

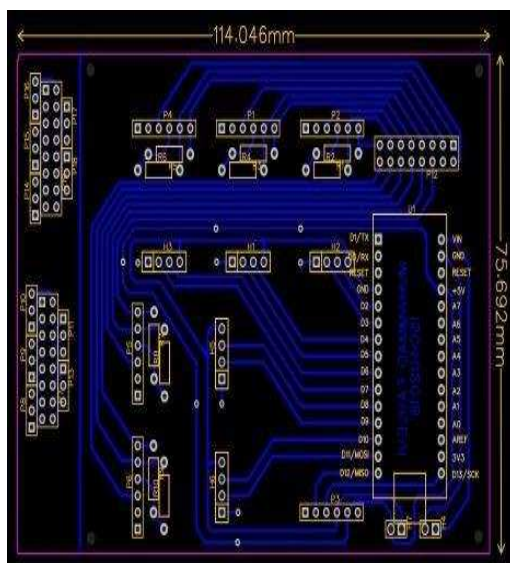


Fig. 2. Ares Proteus

#### G. Sensor Calibration

Before use, the sensor must be calibrated to produce the same value. The value obtained is 2230 for each sensor.

#### H. Sensor placement

The placement of the sensors can be designed as much as possible to get the appropriate coverage of the body load. Two parameters must be determined, namely the number of sensors to be used and how the placement of each sensor on the seat cushion of the chair is to be used.

What becomes a consideration in determining the number and location of sensor placement on the seat cushion is the achievement of coverage from the sensors' reading which can represent data on the tendency of the body position on the object being measured. There must be enough sensors to record enough body weight data received when someone sits. Nevertheless, the number of sensors is not too much to get the efficiency and effectiveness of measurement.

From the experiments conducted, it was determined that the correct number for measuring a person's body weight while sitting was five sensors, assuming this number was sufficient to record body weight data from the lateral (left-right) and anterior (front-back) directions, plus one at the center. The total of these five sensors is expected to be able to get enough body load data while a person is sitting.

Sensor placement must be designed so that all forms of body position with different body load variations can be captured with sensor readings, whether it is an abnormality that is too right or too left or a body position leaning forward or [un leaning backward]. For this reason, the suitable sensor placement is to cover the front-back-right-left and center according to the direction of the wind as proposed in star algorithm [5].

This is by the number of sensors selected: five with four cardinal directions and one in the center. The results of determining the number of sensors and the placement of sensors that have been designed can be seen in Fig. 3 and Fig. 4 below. The sensor composition is then covered with a cushioning board which will distribute the pressure from the body weight when a person sits evenly for the four sensors installed. Thus, it is hoped that the entire sensor can appropriately read the data.

### III. RESULTS AND ANALYSIS

A sitting posture detection system using a force sensor (load sensor) is shown in Fig. 3 and Fig. 4. The outside view of the sitting posture detection system using a force sensor (load sensor) is shown in Fig. 4.

After designing the system, the next thing to do is data collection. The data generated recorded from the sensor is shown in Fig. 5.

After getting five data samples, the next thing to do is data processing. Data processing into *sit mapping* is done through a supporting application (MATLAB). The difference in a person's sitting profile pressure is seen in each sample. The darker the *mapping* sits, the greater the pressure value, as shown in Fig. 6 to Fig. 10.



Fig.3. External view

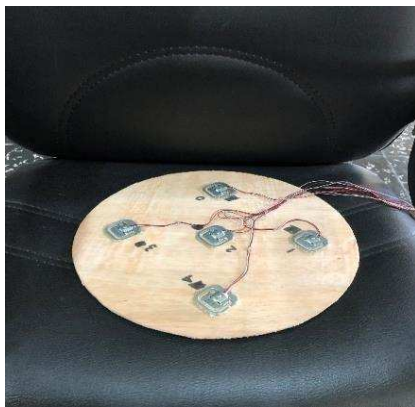


Fig. 4. Inside view

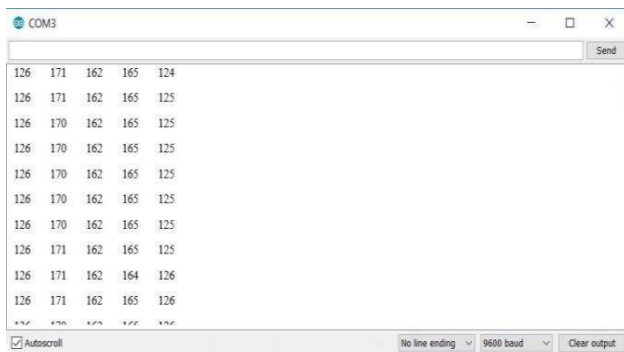


Fig.5. Sensor Result Data Recorded

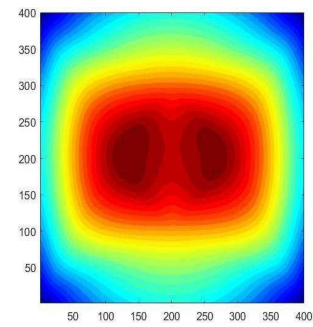


Fig. 6. Sitting Map – Normal Position

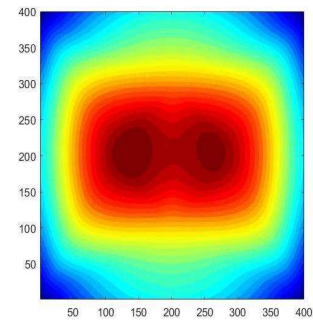


Fig. 7. Sitting Map – Right Leaning Position

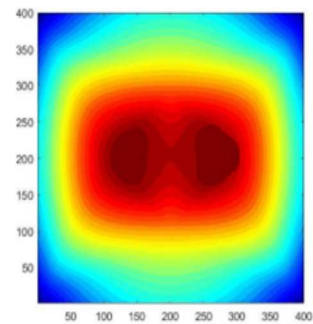


Fig. 8. Sitting Map – Left Leaning Position

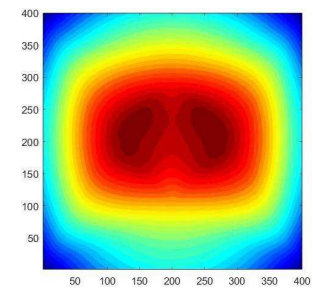


Fig. 9. Sitting Map – Back Leaning Position

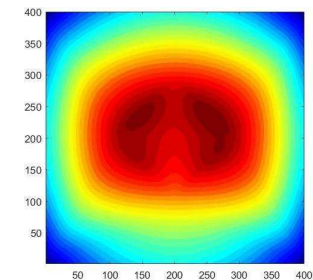


Fig. 10. Sitting Map – Forward Leaning Position

From the samples above, sensor measurements placed in 5 different positions can provide an overview in the form of a sitting map representing the distribution of body weight in a sitting position. This visual form of a heat map can quickly provide information about a person's posture while sitting. The darker the color on the heat map, the greater the reading value of the load sensor. Vice versa, the faded or lighter color on the heat map will show the lower load sensor reading value. The heat map or sitting map will show areas with the same load value and form the appropriate contour. We can project a person's posture or body position while sitting by comparing certain areas on a contour heat map or sitting map. The comparison method can measure which area is more dominant than the left and right or front and back.

Lateral direction can be analyzed with virtual vertical lines that divide the heatmap into two parts: right and left. By comparing the dark red area (which means the area with high load), we can analyze the posture of the body position, whether it is leaning right or left. Likewise, with the comparative method, it can be measured which area is more dominant than either the front or the back. Anterior direction can be analyzed with a virtual horizontal line that divides the heatmap into two parts above and below. By comparing the dark red area (which means the area with a high load), we can analyze the posture of the body position whether it is leaning forward or leaning back.

In sample 1 (Fig. 6), there is a balance of loads both laterally and anteriorly, both in the right-left and front-to-back ratios. Even though it does not exactly show the same area, the difference is still below the acceptable difference threshold, so it can be said that the difference is not too significant. This shows one example of sensor readings indicating a normal body posture position that does not show a right or left leaning and does not show a forward or backward leaning.

In sample 2 (Fig. 7), when compared between the right and left heat map areas, there is a significant difference using the vertical center line. Here it can be seen that the heat map area on the left, which correlates with the right body position, is under more tremendous pressure than the heat map area on the right, which correlates with the left body position. The ratio of the difference in the heatmap area is 3:2, with the left heatmap area being higher than the right. This indicates that the posture of a sitting person is not normal and is more inclined to the right.

On the other hand, in sample 3 (Fig. 8), when compared between the right and left heat map areas, there is a significant difference using the vertical center line. Here it can be seen that the heat map area on the right, which correlates with the left body position, is under more tremendous pressure than the heat map area on the left, which correlates with the right body position. The difference ratio in the heatmap area is 4:5, with the right heatmap area being higher than the right. This indicates that the posture of a sitting person is not normal and is more inclined to the right.

Sample 4 (Fig. 9) and sample 5 (Fig. 10) show a situation where the body's loading conditions are unbalanced between the anterior and posterior parts, as seen from the virtual horizontal line. By comparing the diagonal lines of the heat

map from top to bottom, the tendency of the sitting posture of the position in question can be seen. In Fig. 9, there is an emphasis on the upper part, which indicates a leaning back sitting; in Fig. 10, the emphasis is on the excess weight on the lower part, which indicates a sitting posture leaning forward.

From some of these samples, it can be seen that the ability of the sitting map to produce an analysis of a person's posture and sitting tendency based on the arrangement of sensors and the designed placement follows the needs and answers that need to be resolved.

#### IV. CONCLUSIONS

The conclusions obtained by the study are as follows: Load sensors designed on the system can detect sitting posture properly. Analysis of the data obtained can be used as a diagnostic / examination result because the *sit mapping* of each sample has differences. The more *load* sensors used; the more pressure points can be obtained.

#### ACKNOWLEDGMENT

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